# Appendix A
# Northwest Regional Modeling Consortium
# SFY2027 WORK PLAN

**Project title:** Northwest Regional Prediction Model-2 (WRF)

**Project Duration:** July 1, 2026 – June 30, 2027

**Description and scope of project:**

The Northwest Regional Prediction effort has four major components: 1) high-resolution weather predictions made twice daily using WRF mesoscale atmospheric models, 2) daily ensemble forecasting providing probabilistic and uncertainty information, 3) real-time collection of observations from all available sources throughout the region, and 4) research, analysis, and recommendations for how the modeling system performance can be improved.

**Work Plan:**

The University of Washington will:

1. Provide a suite of deterministic meteorological analyses and forecast products to air quality and land management agencies in Washington, Idaho, and Oregon twice daily, using operational runs from the current and future regional versions of NCEP's Weather Research and Forecasting model (WRF) at 36, 12, 4, and 4/3 km grid resolution. Products are to be available via the University of Washington's Department of Atmospheric Sciences website (http://www.atmos.washington.edu/wrfrt/) at approximately 9:30AM/9:30PM Pacific Standard Time (36, 12, 4 km domains) and approximately 2:30PM/2:30AM (1.3 km grid) following the 1200/0000 UTC initialization times of the modeling system. Forecast products shall extend out to 240 hours for the 36 and 12 km resolutions, 120 hours for the 4 km resolution, and 72 hours for the 4/3 km resolution.

   Provide a 4-km ensemble forecast and web products twice daily at approximately 7:00AM/7:00PM Pacific Standard Time following the 0000 or 1200 UTC initialization times of the modeling system. Ensemble forecast products shall extend out 72 hours.

   This suite of products will be referred to as the University of Washington Environmental Prediction System (ENVPREDSYS). Products shall include:

   a. Surface and upper air charts
   b. Ventilation products

c. Upper air soundings
d. Meteograms
e. Time height series
f. Ensemble plumes
g. Ensemble mean and spread maps

Air quality products, including all the ventilation indices, will be generated as the forecast progresses, rather than when the entire forecast is complete.

2. Maintain and improve the dissemination of the Northwest regional forecasts through the web and other means. Continual improvement will be made to the regional prediction web site, including improved graphics, color schemes, graphical subdomains for the 1.33 km domain, and other additional products. Continue development of ensemble displays, especially the plume products. Implement emergency graphical subdomains and trajectories (e.g., for areas with wildfires), as requested. Specific elements will be determined in collaboration with the NW Regional Modeling Consortium partners who are providing funding. Improvements and additions will be primarily initiated by request of Consortium members. Priority will be based on the relative contributions of participating agencies.

3. Provide periodic assessments of which aspects of model science, if any, are deficient for meteorological modeling needs. Recommend methods for resolving deficiencies and provide written analytical assessments of model science improvements as needed. Make needed improvements in the modeling system, including updating to new versions when verification statistics indicate a positive impact.

4. Provide on-going automated statistical comparisons and verification of the various forecast systems within the ENVPREDSYS. Perform upper-air verification at as many near-surface levels as possible.

   When changes to the model are made, model performance will be assessed. In addition, the changes will be tested for recent periods and locations as provided by Consortium members, or air quality agencies. It is incumbent upon the Consortium members or air quality agencies to provide specifics on these episodes in order for this task to be completed. The procedure for assessment may be included in the Standard Operating Procedure document.

5. Continue assessment of PBL, LSM, and convective schemes in WRF. Assessment will be made for all areas of the domain and times of the year, but particular attention will be paid for performance during wintertime stagnant conditions.

6. Continue to develop and test probabilistic products that have been identified as priorities by Consortium members. Add additional physics members when

possible. Continue to develop the calibration method for the probabilistic forecasts.

7. By request of Consortium members, continue to A) make available to Consortium members by ftp the raw output for the ensemble mean for a specific set of requested variables (~dozen) for a short list of locations, and B) provide by ftp the raw output for the deterministic forecast for a specific set of requested variables (~dozen) for a short list of locations.

8. Make available to Consortium members point extracts of various parameters from different ensemble members, at locations requested, as soon as possible after completion of forecast cycles. Feeds of data files may be requested in different formats (csv, zip etc), and by different transfer methods (eg. ftp, scp).

9. Produce short-term high-resolution model output of localized regions on a case-by-case basis to support NRMC member study of unique air quality or meteorological episodes. Modeling and support to be provided to NRMC members upon special request and only as time and resources allow.

10. To the extent possible with allocated equipment funding: Update (purchase new replacements for) as many as possible of the outdated and out-of-warranty components (e.g. nodes, cores, cables, disks, memory, etc, as needed) of the UW Intel Xeon linux cluster and supporting equipment used to run the ENVPREDSYS. Purchase equipment and supplies for, and maintain, a hard-drive based archival system for data generated by ENVPREDSYS that will replace the former tape-based system. The updates should be optimized to maintain or improve the reliability of the system, and to enable better performance and quicker delivery of the daily products. The components that were replaced may be repurposed to enhance or support additional members in the WRF ensemble system or enhance or support the archival system.

11. Online access to the archival data will be maintained for Consortium members, and will be limited with an access control method decided in consultation with the Consortium. At least the most recent five calendar years (from the current year) of archival data will remain available for Consortium members through this system.

12. Allow Consortium members, with prior approval, to log in to the cluster and run processes with minimal computational and storage requirements, such as WRF to HYSPLIT.

13. Maintain generating BUFKIT files for a predetermined set of locations with each twice-daily model run. Consortium members may request adding locations to the predetermined set.

14. Allow Consortium members, with prior approval, to provide and bring or send a physical hard drive to the UW facilities and download WRF output files.

15. Continue to monitor for web access and download activity that might hinder the performance of the ENVPREDSYS, hinder access to ENVPREDSYS products for Consortium members, or provide significant financial benefit for a non-member. Take steps to block or reduce access for any users that might be doing so, by blocking IPs or implementing other controls as deemed necessary, with consultation with the Consortium.

16. Maintain a suite of enhanced graphics behind an access wall.

17. Investigate and report on the requirements and consequences of using the ECMWF AIFS or AFS global model to drive the ENVPREDSYS. Identify any potential problems such as, but not limited to: grid and temporal resolution, parameter mismatches, bias and failure modes, reliability, and the resulting skill when used to drive the ENVPREDSYS.

**Deliverables:**

1. ENVPREDSYS products will be produced and updated to the website 365 days a year with a reliability measure of at least 97%.

2. Provide ENVPREDSYS gridded data fields from the 00 UTC operational runs to Washington State University-AIRPACT prediction system by 08 UTC.

3. Annual "state of the model" report must be provided to Project Sponsor for the contract period ending June 30, 2027. This report can be in the form of a presentation to the Consortium. The report must detail progress made against specific deliverables, status of performance measures, problems encountered and or resolved, and future direction/recommendations. Include status of ENVPREDSYS work including progress on goals, specific deliverables, and status as to performance measures.

**What is the environmental outcome?** This project provides for the underlying data needed to drive a suite of air quality prediction and forecasting tools. These tools are used for forecasting air quality in the Pacific Northwest and also for assessing the impacts to air quality resulting from direct emission reduction activities (e.g. burn bans, smoke management, daily air quality forecasts). As a result, this project provides critical data essential to air quality planning and daily operations of air quality management programs. These activities will result in improved air quality as well as improved warnings to allow protection of vulnerable populations.

Project Period: July 1, 2026 through June 30, 2027
Project Sponsor: Lance Giles, Lane Regional Air Protection Agency
Project lead or technical lead/number: Dr. Clifford Mass, University of Washington